

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS
NANYANG TECHNOLOGICAL UNIVERSITY

Review Questions with solutions

Part 2

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 These are review questions for Chapter 4 in Ross' book.

- (1) If X is a binomial random variable with expected value 6 and variance 2.4, find $P\{X = 5\}$.

Solution: Since $\mathbb{E}(X) = np$, $\text{Var}(X) = np(1-p)$, we are given that $np = 6$, $np(1-p) = 2.4$. Thus, $1-p = 0.4$, $p = 0.6$, $n = 10$. Thus,

$$P\{X = 5\} = \binom{10}{5} (0.6)^5 (0.4)^5.$$

- (2) Show that if X is a geometric random variable with parameter p , then

$$\mathbb{E}(1/X) = \frac{-p \log(p)}{1-p}.$$

HINT: You will need to evaluate an expression of the form $\sum_{i=1}^{\infty} a^i/i$. To do so, write a^i/i as $\int_0^a x^{i-1} dx$, and then interchange the sum and the integral.

Solution: We let $q = 1-p$. Then

$$\begin{aligned} \mathbb{E}\left(\frac{1}{X}\right) &= \sum_{n=1}^{\infty} \frac{1}{n} q^{n-1} p = \frac{p}{q} \sum_{n=1}^{\infty} \frac{q^n}{n} \\ &= \frac{p}{q} \sum_{n=1}^{\infty} \int_0^q x^{n-1} dx = \frac{p}{q} \int_0^q \left(\sum_{n=1}^{\infty} x^{n-1} \right) dx \\ &= \frac{p}{q} \int_0^q \frac{1}{1-x} dx = \frac{p}{q} \int_p^1 \frac{1}{z} dz = -\frac{p}{q} \log p. \end{aligned}$$

- (3) Suppose that

$$P\{X = a\} = p, \quad P\{X = b\} = 1-p.$$

- (a) Show that $\frac{X-b}{a-b}$ is a Bernoulli random variable.
 (b) Find $\text{Var}(X)$.

Solution: Since $\frac{X-b}{a-b}$ will equal 1 with probability p or 0 with probability $1-p$, it follows that it is a Bernoulli random variable with parameter p . Because the variance of such a Bernoulli random variable is $p(1-p)$, we have

$$p(1-p) = \text{Var}\left(\frac{X-b}{a-b}\right) = \frac{1}{(a-b)^2} \text{Var}(X-b) = \frac{1}{(a-b)^2} \text{Var}(X),$$

and $\text{Var}(X) = p(1-p)a - b^2$.

- (4) Each game you play is a win with probability p . You plan to play 5 games, but if you win the fifth game, then you will keep on playing until you lose.

- (a) Find the expected number of games that you play.
- (b) Find the expected number of games that you lose.

Solution: Let X denote the number of games that you play and Y the number of games that you lose.

- (a) After your fourth game, you will continue to play until you lose. Therefore, $X - 4$ is a geometric random variable with parameter $1 - p$, so

$$\mathbb{E}(X) = \mathbb{E}(4 + (X - 4)) = 4 + \mathbb{E}(X - 4) = 4 + \frac{1}{1 - p}.$$

- (b) We let Z denote the number of losses you have in the first 4 games. Then Z is a binomial random variable with parameters 4 and $1 - p$. By $Y = Z + 1$, we have

$$\mathbb{E}(Y) = \mathbb{E}(Z + 1) = \mathbb{E}(Z) + 1 = 4(1 - p) + 1.$$

- (5) Ten balls are to be distributed among 5 urns, with each ball going into urn i with probability p_i , $\sum_{i=1}^5 p_i = 1$. Let X_i denote the number of balls that go into urn i . Assume that events corresponding to the locations of different balls are independent.
- (a) What type of random variable is X_i ? Be as specific as possible.
 - (b) For $i \neq j$, what type of random variable is $X_i + X_j$?
 - (c) Find $P\{X_1 + X_2 + X_3 = 7\}$.

Solution:

- (a) Because each ball independently goes into urn i with the same probability p_i , it follows that X_i is a binomial random variable with parameters $n = 10$, $p = p_i$.
- (b) First note that $X_i + X_j$ is the number of balls that go into either urn i or urn j . Then, because each of the 10 balls independently goes into one of these urns with probability $p_i + p_j$, it follows that $X_i + X_j$ is a binomial random variable with parameters 10 and $p_i + p_j$.
- (c) By the same argument, $X_1 + X_2 + X_3$ is a binomial random variable with parameters 10 and $p_1 + p_2 + p_3$. Therefore,

$$P\{X_1 + X_2 + X_3 = 7\} = \binom{10}{7} (p_1 + p_2 + p_3)^7 (p_4 + p_5)^3.$$

- (6) Customers arrive at a shop randomly with a rate of 4 per hour, modelled by a Poisson process. In a one-hour period, calculate the conditional probability that no customers arrive in the second half hour given that
- (a) exactly 4 customers arrive in the first half hour;
 - (b) exactly 4 customers arrive over the whole one-hour period.

Solution:

- (a) Let X be the number of customers who arrive in the second half hour, then $X \sim \text{Poisson}(2)$. By independence with the customers arriving in the first half hour,

$$P\{X = 0\} = e^{-2}.$$

- (b) Let Y be the number of customers who arrive in the first half hour and N be the number of customers who arrive over the one-hour period, then $Y \sim \text{Poisson}(2)$

and $N \sim \text{Poisson}(4)$.

$$\begin{aligned}
 P\{X = 0|N = 4\} &= \frac{P\{X = 0 \text{ and } N = 4\}}{P\{N = 4\}} \\
 &= \frac{P\{X = 0\}P\{Y = 4\}}{P\{N = 4\}} \\
 &= \frac{e^{-2} \cdot \frac{2^4}{4!} e^{-2}}{\frac{4^4}{4!} e^{-4}} \\
 &= \frac{1}{16}.
 \end{aligned}$$

(7) Let X be a Poisson random variable with parameter λ . Prove that

$$\mathbb{E}(X^n) = \lambda \mathbb{E}((X + 1)^{n-1}).$$

Now use this result to show that $\mathbb{E}[X^3] = \lambda^3 + 3\lambda^2 + \lambda$.

Proof:

$$\begin{aligned}
 \mathbb{E}(X^n) &= \sum_{k=1}^{\infty} k^n e^{-\lambda} \frac{\lambda^k}{k!} \\
 &= \sum_{k=0}^{\infty} (k+1)^n e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!} \\
 &= \lambda \sum_{k=0}^{\infty} (k+1)^{n-1} e^{-\lambda} \frac{\lambda^k}{k!} \\
 &= \lambda \mathbb{E}[(X + 1)^{n-1}].
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \mathbb{E}(X^3) &= \lambda \mathbb{E}((X + 1)^2) = \lambda \mathbb{E}(X^2 + 2X + 1) \\
 &= \lambda(\lambda(\lambda + 1) + 2\lambda + 1) = \lambda^3 + 3\lambda^2 + \lambda.
 \end{aligned}$$